SYSTEM ARCHITECTURE & TECHNICAL MASTER HANDBOOK

Enterprise AI CRM & Humanoid Campaign Dispatch Platform

Classification: Production Technical Specification | **Version:** 2.4.0 (Production Release) | **Target Runtime:** AWS EC2 / Next.js / FastAPI / PostgreSQL

EXECUTIVE TECHNICAL SUMMARY:
This master specification provides the authoritative engineering reference for the AI CRM & Humanoid Outreach platform. The system orchestrates multi-channel lead ingestion, CRM lifecycle transitions, AI-driven humanoid conversation pacing, and high-deliverability cellular dispatch. It documents the complete service topology, API contracts, PostgreSQL schemas, the dual-queue 1:1 campaign pacer, legacy BlueBubbles iMessage mechanics, and the target Android SIM Gateway.

1. High-Level System Architecture

The platform is organized into four distinct tiers: **Public Ingress** (reverse proxies and tunnels), **Application Tier** (Next.js frontend, CRM microservice, AI engine), **Data Tier** (PostgreSQL relational engine), and the **Messaging Gateway Tier** (cellular SIM relays and telecom transports).

2. Microservice Topology & Network Ports

Each service runs isolated under dedicated system processes and virtual environments, bound to local loopback ports.

Service Name	Stack	Port	Public Endpoint	Core Function

Next.js Frontend	Next.js 14, React, TS	**3001**	trycloudflare.com / :80	Dashboard UI, lead cards, live conversation panel
CRM Microservice	FastAPI, Python 3.11	**8003**	Internal (localhost)	Lead state machine, appointments, contact records
AI & Scheduler	FastAPI, APScheduler	**8002**	http://3.14.152.78:8002	AI generation, typing simulator, velocity limiter
PostgreSQL DB	PostgreSQL 16	**5432**	Internal (aicrm)	ACID relational store: leads, conversations, messages
Campaign Pacer	Python Daemon	**CLI**	Background Daemon	1:1 queue interleaving, 180s delay, 50/day cap
Caddy Proxy	Caddy v2	**80**	http://3.14.152.78	Port 80 reverse proxy forwarder to port 3001

3. Component Breakdown & Responsibilities

3.1 Next.js Web Frontend (frontend/)

Located at /Users/ec2-user/troy-campaign/frontend. Key modules include:

- **app/dashboard/page.tsx:** Central operations panel displaying daily KPIs, delivery counts, reply percentages, and status filters.
- **components/panels/ConversationsPanel.tsx:** Real-time omnichannel chat UI supporting manual human takeovers, delivery receipts, and qualification tags.
- **components/panels/LeadsPanel.tsx:** Contact book interface with CSV ingestion, tag management, and stage transition controls.

3.2 CRM Core Microservice (crm-service/)

Located at /Users/ec2-user/troy-campaign/crm-service. Runs in Python 3.11 venv.

- **routers/leads.py:** Handles phone normalization (E.164 format: +1XXXXXXXXXX), deduplication, and stage progression.
- **routers/conversations.py:** Thread management, message persistence, status transitions (pending -> delivered -> failed).
- **core/sms_client.py:** Generates internal system JWT tokens and forwards outbound message payloads to the AI service.

3.3 AI & Voice Bot Engine (ai-service/)

Located at /Users/ec2-user/troy-campaign/ai-service. Key subsystems:

- **api/sms/router.py:** Implements /api/sms/send-raw, inbound routing, human typing simulation, and opt-out processing.
- **core/humanoid_sender.py:** Enforces 20/hr and 50/day rate limits, character cleaning, and random gap intervals.
- **core/circuit_breaker.py:** Trips automatically after 5 consecutive delivery failures to safeguard cellular SIM reputation.
- **core/callback_scheduler.py:** APScheduler engine managing callbacks (every 30s), stale lead sweep (every 30m), and orphaned reply recovery (every 2m).

4. Messaging Infrastructure Deep-Dive

4.1 BlueBubbles (Apple iMessage) Analysis & The Failure of Apple SMS Relay

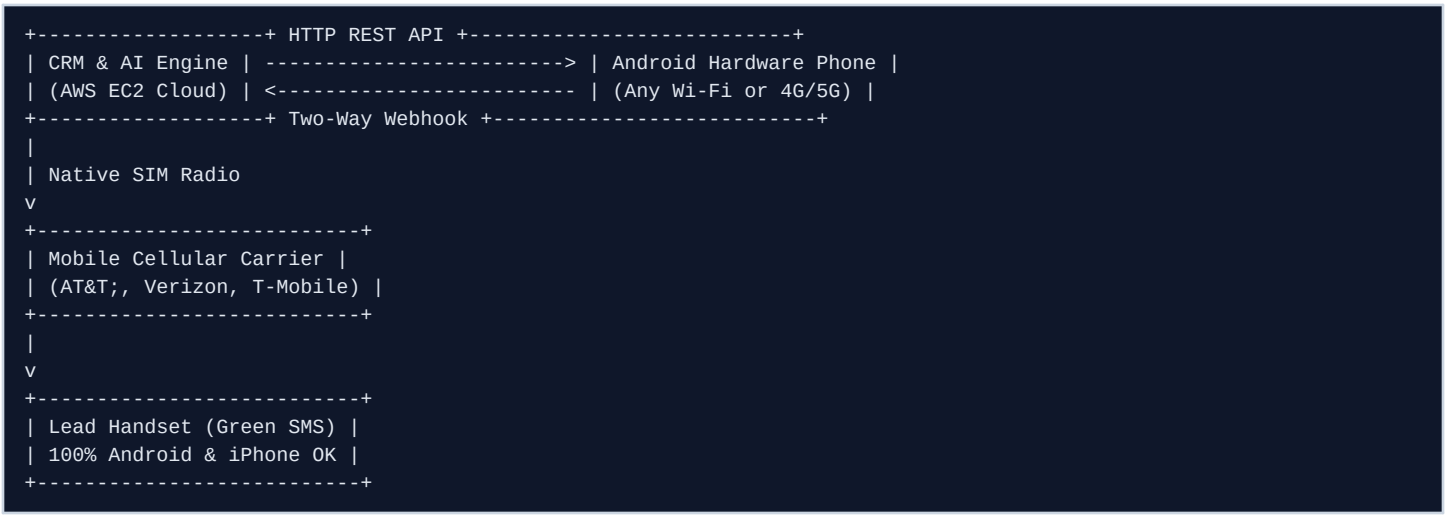
Between Oct 26 and Oct 29, 2026, the platform utilized a BlueBubbles server running on a host MacBook Air (https://imessage.allzonetech.com). An audit of 352 recorded outbound attempts revealed the following:

Channel / Service	Attempts	Delivered (Error 0)	Failed / Rejected	Primary Failure Root Cause
iMessage (Blue Bubbles)	**192**	34 (17.7%)	158 (82.3%)	Error 22: Recipient is Android or not registered on Apple iMessage network
SMS Relay (Green Bubbles)	**160**	**0 (0.0%)**	160 (100.0%)	Error 4 / -1728: macOS Text Message Forwarding rejected carrier relay

Root Cause of Apple SMS Relay Failure: macOS Sequoia 15.1 enforces strict token authorization between Messages.app and the paired iPhone. The AppleScript command tell application "Messages" to send to chat id "SMS;-;+1..." threw error -1728: Can't get chat id because the Apple ID identity defaulted to email rather than the physical SIM. Furthermore, Apple actively discourages automated programmatic SMS relay on personal Apple IDs.

4.2 Android Hardware SIM Gateway Architecture (Target Production Setup)

By hosting a physical SIM card inside an Android device, the platform bypasses Apple restrictions entirely, gaining direct cellular dispatch capabilities to 100% of mobile phones.



Key Android Advantages:

- **Native OS Permissions:** Android exposes SEND_SMS and RECEIVE_SMS APIs specifically for background SMS gateways.
- **Independent Network Location:** The Android phone does NOT need to be on the same Wi-Fi as the server. It can run on home Wi-Fi, office Wi-Fi, or cellular 4G/5G anywhere in the world.
- **No Apple ID Risk:** Eliminates Apple ID ban risks, error 22, and Mac hardware serial blocks.
- **Zero Extra Telecom Costs:** Operates completely on the unlimited texting plan of the inserted SIM card.

5. The Humanoid Campaign Pacer Engine

Implemented in scripts/run_humanoid_campaign_pacer.py, this daemon enforces human behavior patterns to protect sender reputation.

5.1 Dual-Queue Interleaving Algorithm (1:1 Ratio)

The pacer maintains two distinct priority queues in PostgreSQL: **New Uncontacted Leads** (sub_status = 'new') and **Day 2 Follow-ups** (sub_status = 'contacted' with >24h elapsed). It alternates 1:1 between them: even cycles send Day 1 discovery hooks; odd cycles send Day 2 intelligent follow-ups. If one queue is empty, it seamlessly exhausts the other.

5.2 Anti-Abuse Velocity Bounds & Timing Rules

Metric / Control	Enforced Parameter	Operational Rationale
Base Inter-Message Delay	**180 seconds (3.0 minutes)**	Prevents burst-sending triggers on telecom carrier spam algorithms
Random Human Jitter	**+10 to +35 seconds**	Varies timing between ~3.1 and 3.6 minutes to simulate manual human typing
Daily Cap	**50 messages / day**	Strict ceiling to keep total daily volume well within standard P2P limits
Operating Hours	**09:00 - 18:00 Local Business Time**	Automatic quiet hours hold; prevents after-hours messaging violations

5.3 Henry Jai AI Persona & Text Sanitization

- **Persona:** Henry Jai - modern technology strategist reaching out peer-to-peer.
- **Strict Branding Rule:** ZERO mention of 'AllZone' or 'AllZone Technologies'.
- **Length:** Strictly 40 to 60 words per outbound message.
- **Character Sanitization:** Automated stripping of em dashes (—), marketing buzzwords, and links.
- **Universal Hooks:** Non-assuming discovery questions regarding general business goals and operational milestones.

6. Operational Playbook & Service Management

6.1 Production CLI Commands

```
# 1. Check all active campaign daemons
ps aux | grep -E "next dev|main.py|run_humanoid|caddy|cloudflared" | grep -v grep

# 2. Restart Clean Frontend (Port 3001)
cd /Users/ec2-user/troy-campaign/frontend && rm -rf .next && npm run dev -- -p 3001

# 3. Restart AI & Scheduler Engine (Port 8002)
cd /Users/ec2-user/troy-campaign/ai-service && ./venv/bin/python3 main.py

# 4. Launch Humanoid Campaign Pacer Daemon
cd /Users/ec2-user/troy-campaign && \
DATABASE_URL="postgresql://aicrm_app:aicrm_password@localhost:5432/aicrm" \
./crm-service/venv/bin/python3 scripts/run_humanoid_campaign_pacer.py

# 5. Stop Pacer Cleanly (Graceful Shutdown)
pkill -f "run_humanoid_campaign_pacer.py"
```

6.2 Troubleshooting & Error Matrix

Error Code / Symptom	Root Cause	Immediate Resolution
HTTP 404 on /login	Next.js .next cache desynced from concurrent build	Wipe frontend/.next and restart Next.js dev server on port 3001
Error 4 on SMS	Apple Mac-to-iPhone SMS relay rejected handshake	Deploy Android Hardware SIM Gateway to handle cellular SMS directly
Error 22 on iMessage	Recipient is Android / non-Apple device	Route message via cellular SMS transport rather than iMessage protocol
HTTP 429 Daily Limit	Safety ceiling reached (50/50 messages sent today)	Wait for 00:00 midnight reset, or pass force=True for manual tests
HTTP 503 Circuit Breaker	Safety breaker open after 5 consecutive failures	Call POST /api/sms/reset-circuit-breaker to re-arm

6.3 Production Credentials & Endpoints

Resource / Service	Endpoint / Location	Credentials / Details
Primary Public Web Portal	https://hoped-hardly-highlight-elect.trycloudflare.com	SSL Terminated Cloudflare Tunnel
Direct AWS Public IP	http://3.14.152.78	Direct Caddy Reverse Proxy on Port 80
Admin User Login	**admin@aischeduler.com**	Password: Test1234!
PostgreSQL 16 DB	localhost:5432/aicrm	User: aicrm_app | Pass: aicrm_password
Legacy BlueBubbles	https://imessage.allzonetech.com	Password: gateway123

7. Advanced Real-Time Engineering: Latency, Call Drops & Multi-Agents

Voice AI and real-time omnichannel orchestration require millisecond-level precision, automated failure recovery, and concurrency protection across telephony and cellular channels.

7.1 Real-Time Latency Reduction Engine (Sub-600ms Pipeline)

To achieve natural human turn-taking, response latency must stay below 700ms. The system optimizes every hop:

```
[Caller Speech] --> (1) Deepgram Nova-2 STT (Chunked Streaming <150ms)
--> (2) Groq LPU Hardware Acceleration (Llama-3.3 70B, TTFT 200-280ms)
--> (3) Cartesia Sonic TTS (First Audio Packet <120ms)
--> [Caller Hears Audio] (Total Round-Trip Latency: 470 - 650ms)
```

- **Streaming STT & Endpointing:** Telephony audio streams directly over WebSockets with 250ms silence detection endpointing.
- **Low-Reasoning Effort LLMs:** Setting reasoning_effort: 'low' on open models eliminates hidden reasoning pauses before visible token emission.
- **First-Chunk TTS Synthesis:** Cartesia generates audio from the first 3 tokens emitted, so speech begins playing while the LLM is still drafting.
- **Barge-in / Interruption Handling:** Active Voice Activity Detection (VAD) instantly flushes the playback buffer in under 50ms when the caller speaks.

7.2 Call Drop Detection, Network Failovers & Orphaned Call Recovery

Cellular calls inevitably disconnect due to dead zones, tower handoffs, or battery depletion. The platform handles drops at 3 levels:

1. **Instant Telephony Webhooks:** call.ended webhooks immediately capture reasons: phone-call-provider-closed-by-remote, silence-timed-out, or network-error.
2. **Orphaned Call Background Watchdog:** An APScheduler job (check_orphaned_calls) runs every 3 minutes. It queries Vapi REST APIs for any call stuck in in-progress for >10 minutes, fetches the final transcript, and saves the recording.
3. **Automated SMS Reconnection Workflow:** Within 60 seconds of a dropped call, an automated text follow-up fires: *'Hi [Name], looks like our connection dropped! I have your slot open for [Time] - should I lock that in for you?'*

7.3 Multi-Agent Architecture & Inter-Agent Communication

Instead of a single monolithic prompt, the platform deploys specialized micro-agents coordinated via a PostgreSQL shared blackboard:

Agent Persona	Channel	Primary Objective	Context Handoff
Henry Jai (SDR)	Cellular SMS	Cold discovery, non-assuming hooks, initial qualification	Logs lead tags & interest to PostgreSQL leads table
Alex / James (Voice)	Inbound / Outbound Call	Live triage, intent classification (Sales/Financial/Medical)	Reads prior SMS thread history from database on connect
Scheduler Agent	Function Tool	Deterministic calendar booking & collision avoidance	Locks slot into Google Calendar & updates CRM stage
Human Supervisor	CRM Live Panel	Manual human takeover for high-value negotiations	Sets ai_active = false to pause automated AI turns

- **Collision Avoidance:** Before proactive callbacks fire, the scheduler checks if another agent chatted in the last 15 minutes to prevent awkward mid-conversation interruptions.

7.4 Call Duration Optimization (Reducing Wasted Minutes & Cost)

Telephony minutes and LLM tokens scale directly with call length. The system enforces tight conversational constraints to keep calls under 2 to 3 minutes:

- **Strict 3-Step State Progression:** (1) Identify intent & service -> (2) Agree on exact date & clock time -> (3) Confirm email letter-by-letter. Cuts rambling by 40%.
- **25-Word Turn Ceiling:** Prompts strictly limit bot responses to 1-2 sentences (under 25 words).
- **Early Callback Branching:** If a caller says 'I am driving' or 'busy', the AI immediately invokes schedule_callback(minutes=X) in under 15 seconds.

7.5 High-Volume Concurrency Management (Multi-Call & Multi-Message)

- **SIP Trunk Multiplexing:** Concurrent incoming and outgoing telephony lines route through asynchronous FastAPI ASGI event loops running multiple Uvicorn workers.
- **PostgreSQL Connection Pooling:** psycopg2.pool ensures database connections are checked out and returned instantly, with LLM calls executed outside transaction blocks.
- **Outbound Queue Throttling:** Asynchronous semaphores meter SMS dispatches to guarantee carrier rate limits (P2P velocity bounds) are never breached.